Technology Stack

Date	15 July 2026
Team ID	*************
Project Name	SmartLender – Applicant Credibility Prediction for Loan Approval
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Jinja2 Templates	Used to build a simple and responsive web interface for entering applicant details and displaying the prediction result.
2	Backend / Server-Side	Python 3, Flask	Flask handles user requests, processes input data, loads the trained machine learning model, and returns the prediction.
3	Machine Learning	Scikit-learn (RandomForestClassifier), Pandas, NumPy, SMOTE, StandardScaler	Pandas and NumPy are used for data manipulation, SMOTE balances the dataset, StandardScaler scales features, and Random Forest predicts loan approval.
4	Database / Data Storage	CSV Dataset (loan_prediction.csv), Pickle Model (smart_lender_model.pkl)	The CSV file is used for training the model, while the Pickle file stores the trained model for prediction during runtime.
5	Development Environment	Google Colab, Visual Studio Code (or any Python IDE)	Google Colab was used for model training and experimentation. VS Code or another IDE can be used to develop and run the Flask application.

6	Version Control	Git & GitHub	Used to maintain project versions, collaborate with team members, and host the project repository.
7	Deployment / Hosting	Flask Local Development Server	The application runs locally using python app.py, making it easy to demonstrate during the internship.